

PROJECT REPORT

Sales Data Analysis

European Supermarket Sales Dashboard

Data analysis with Excel, Power BI and Python

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Abstract

This project analyses the sales of a European supermarket business using a dataset of 4,117 orders and 8,045 product lines placed between 2017-01-01 and 2020-12-31 across 15 countries. The data was first summarised with pivot tables and slicers in Microsoft Excel, then modelled and visualised in an interactive Power BI dashboard, and finally re-analysed in Python to check the dashboard figures and to look for patterns that the dashboard does not show.

The business generated total sales of 2.35M and a profit of 283,202, a profit margin of 12.1%. Sales grew from 414,493 in 2017 to 755,030 in 2020 (22.1% a year), but profit grew more slowly and the margin slipped from 13.2% to 11.3%. Technology is the largest category by sales, while Office Supplies earns the highest margin (15.2%) and the most profit.

The most important finding is about discounting. Order lines with a discount of 30% or more make up 17.7% of all lines but lose 143,852 in total. Without them, the remaining lines earn a margin of 21.0%. Five countries (Netherlands, Sweden, Ireland, Portugal and Denmark) sell almost entirely at about 50% discount and are all loss-making. The report documents how every figure on the dashboard reconciles to the source data, evaluates the dashboard design, and recommends changes to both the business and the dashboard.

Keywords: sales analysis, Power BI, Excel pivot tables, dashboard, profit margin, discount analysis, data visualisation

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1. Introduction

1.1 Background

Retail businesses record every sale, but raw transaction tables are hard to read. Managers need to know how sales and profit are changing, which products and markets drive results, and where money is being lost. Dashboards answer these questions at a glance, and a good dashboard is backed by careful analysis of the data underneath it.

This project follows that full path. It starts from a supermarket sales workbook, builds summaries in Excel, creates a one-page Power BI dashboard titled "Sales Data Dashboard", and then goes beyond the dashboard with a deeper analysis in Python.

1.2 Problem statement

The business has four years of order data from 15 European countries but no single view of how it is performing. The problem is to turn the data into a clear, interactive dashboard, to verify that the dashboard shows the right numbers, and to find out what the data says about growth, profitability, discounts and returns.

1.3 Objectives

- Clean and understand the two-table sales dataset and its relationship.
- Summarise the data in Excel with pivot tables, charts and slicers.
- Build an interactive Power BI dashboard showing the main sales and profit measures.
- Verify every dashboard figure against the source data.
- Analyse sales and profit by time, category, sub-category, product, country, region, customer segment and ship mode.
- Measure the effect of discounts and returns on profit.
- Evaluate the dashboard and recommend improvements and business actions.

1.4 Scope

The analysis covers the single dataset supplied, from January 2017 to December 2020. The file does not state the currency, so values are reported as plain monetary amounts without a currency symbol. Cost and inventory data are not available, so profit is taken as recorded in the file.

2. Tools and Technologies

Table 1: Tools used

Tool	Role in the project
Microsoft Excel	Source workbook; pivot tables, charts and slicers for first-level summaries
Microsoft Power BI Desktop	Data model, DAX-ready measures, interactive dashboard (bridge_group.pbix)
Python (Pandas, NumPy)	Joining the tables, cleaning checks, aggregations and KPI reconciliation
Matplotlib and Seaborn	Charts used throughout this report
Jupyter Notebook	Reproducible record of the analysis (European_Sales_Analysis.ipynb)

Power BI is the main delivery tool for the business user, while Python is used here as an independent check. Because the same totals appear in both, the Python results give confidence that the Power BI model, relationships and visuals are correct.

3. Dataset Description

3.1 Structure

The workbook `European_supermarket_sales_data.xlsx` holds two data tables. List of Orders has one row per order (4,117 rows, 12 columns) and describes who ordered, where, when and how it was shipped. Order Breakdown has one row per product line (8,045 rows, 9 columns) and holds the sales, profit, discount, quantity, category and return status. The tables are linked by Order ID. Two further sheets (Sheet1 and Sheet2) contain Excel pivot tables built on these tables.

Table 2: Columns of the two data tables

Table	Column	Meaning
List of Orders	Order ID	Unique order number (key linking both tables)
	Order Date, Ship Date	Dates the order was placed and shipped
	Customer Name	Customer who placed the order
	City, State, Country	Delivery location (15 countries)
	Region	Central, North or South
	Segment	Consumer, Corporate or Home Office
	Ship Mode	Economy, Economy Plus, Priority or Immediate
	lon, lat	Map coordinates of the delivery city
Order Breakdown	Product Name	Product sold on the line
	Category, Sub-Category	3 categories, 17 sub-categories
	Sales, Profit, Quantity	Value, profit and units of the line
	Discount	Discount rate applied (0 to 0.8)
	Return Status	Sold or Returned

3.2 Data quality checks

Table 3: Data quality summary

Check	Result
Missing values (both tables)	0
Duplicate rows (both tables)	0
Order breakdown lines without a matching order	0
Unique orders / customers / products	4,117 / 792 / 1,810
Period covered	2017-01-01 to 2020-12-31
Orders with ship date before order date	24 (0.58%)

The data is clean: there are no missing values or duplicates, and every product line matches an order, so a join on Order ID loses nothing. The only anomaly is that 24 orders show a ship date earlier than the order date, by as much as 365 days, which looks like a data-entry error (probably a wrong year). This

does not affect sales or profit analysis and these rows were kept, but any delivery-time analysis would need them corrected first.

3.3 Preparing the data

For analysis the two tables were joined on Order ID into one table of product lines, which gives each line its date, country, region, segment and ship mode. In Power BI the same link is a one-to-many relationship from List of Orders to Order Breakdown. Year and month were derived from the order date. No rows were removed.

4. Methodology

1. Load both tables and check data quality (Section 3).
2. Summarise the data with Excel pivot tables, charts and slicers (Section 5).
3. Build the Power BI model and dashboard (Section 6).
4. Recompute every dashboard number in Python from the source file and compare (Section 6.4).
5. Analyse trends and segments in more depth than the dashboard allows (Section 7).
6. Evaluate the dashboard and summarise insights, recommendations and limitations (Sections 8 to 11).

Three measures are used throughout. Sales is the value of the product lines, Profit is the profit recorded for them, and Profit margin is Profit divided by Sales. Where discount is analysed, order lines are grouped by their discount rate. Percentages of lines mean the share of the 8,045 product lines, not of orders.

5. Excel Work

5.1 Pivot tables, charts and slicers

The Excel workbook contains two pivot tables, each with a chart, and a set of slicers for interactive filtering.

Table 4: Pivot tables in the Excel workbook

Sheet	Pivot table	Chart	Purpose
Sheet1	Order Date by year; Sum of lon and Sum of lat	Bar chart	Yearly summary of the location fields
Sheet2	Order ID; Sum of Sales, Sum of Discount, Sum of Profit	Line chart	Sales, discount and profit by order

Seven slicers sit on Sheet2: Order ID, Product Name, Discount, Profit, Quantity, Sub-Category and Return Status. They let a user filter the pivot table with one click, for example to see only returned items or only one sub-category.

5.2 Observations

Pivot tables are a quick way to explore data, and the slicers make them interactive. Two limits became clear while reviewing the workbook. First, summing longitude and latitude by year (Sheet1) does not produce a meaningful business measure, because coordinates are labels for places and not quantities. A count of orders or the sum of Sales by year would carry real information. Second, the chart on Sheet2 draws one point per order, so with thousands of orders it is hard to read. Summarising by year, month, category or country would give a clearer line.

These limits are part of the reason the project moves to Power BI, where the same data is modelled once and then explored through linked visuals, a map and cross-filtering slicers.

6. Power BI Dashboard

6.1 Dashboard overview

The dashboard, saved as `bridge_group.pbix`, has a main page titled "SALES DATA DASHBOARD" on a 1280 by 720 canvas, shown in Figure 1. It uses two tables, List of Orders and Order Breakdown, related on Order ID.

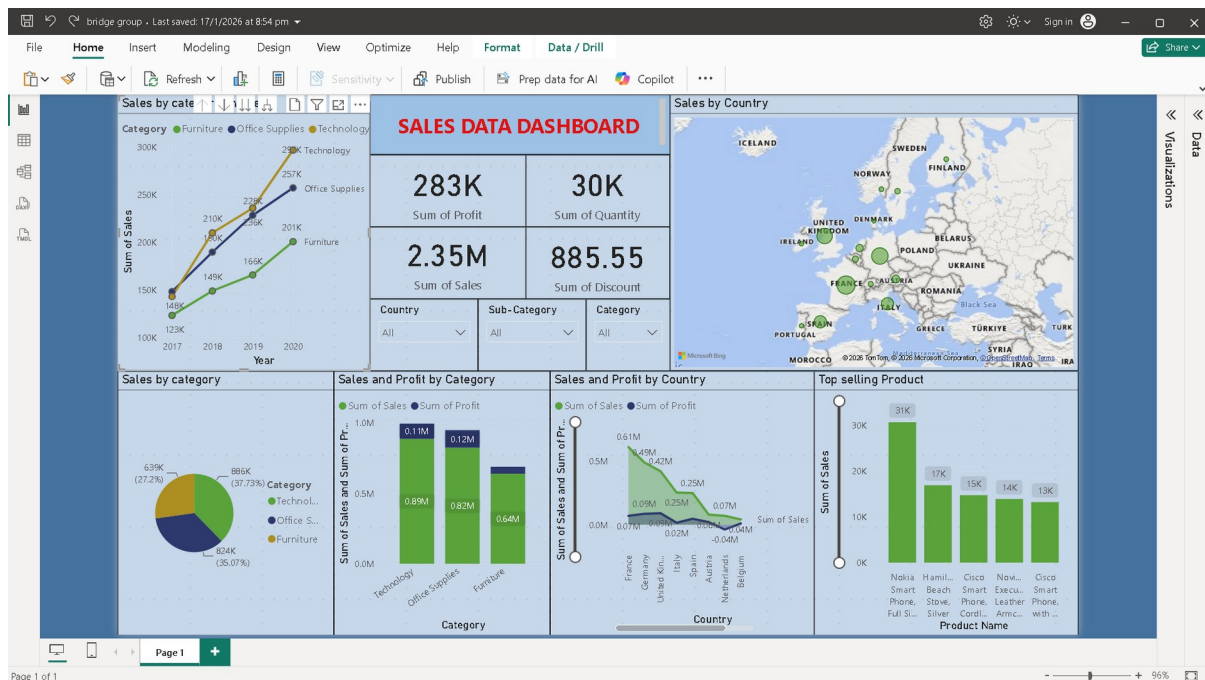


Figure 1: Power BI dashboard (Page 1)

6.2 Visuals

Table 5: Visuals on the dashboard page

Visual	Type	Fields	Answers
Profit, Quantity, Sales, Discount	Cards (4)	Sum of each measure	Headline totals
Sales by category	Line chart	Year, Category, Sales	How each category grew
Sales sum by Country	Map	Country, Sales	Where sales happen
Sales by category	Pie chart	Category, Sales	Category share
Sales and Profit by Category	Stacked column	Category, Sales, Profit	Sales vs profit per category
Sales and Profit by Country	Area chart	Country, Sales, Profit	Country performance
Top selling Product	Column chart	Product Name, Sales	Best-selling products
Country, Sub-Category, Category	Slicers (3)	Dropdown filters	Interactive filtering

6.3 Interactivity

All visuals are linked. Choosing a country, category or sub-category in a slicer, or clicking a bar, a pie slice or a map bubble, filters every other visual on the page. The line chart is built on the date hierarchy (year, quarter, month, day), so it can be drilled down from yearly to daily sales. This lets one page answer many questions without extra reports.

The .pbix file I received also holds a second report page. From its layout it contains four cards (Sales, Quantity, Profit, Discount), a clustered column of Sales by Category and Sub-Category, a map, a pie chart, a Sub-Category slicer, a line chart of Sales over the date hierarchy and a table of Sales by Country and Category. It was not part of the screenshot, so it is described here only from its field list.

6.4 Reconciliation of dashboard figures

Every number visible on the dashboard was recomputed in Python from the source Excel file. All of them match, which confirms that the relationship, the aggregations and the visuals are correct.

Table 6: Dashboard values compared with the source data

Dashboard figure	Dashboard	Recomputed from data	Match
Sum of Profit	283K	283,202	Yes
Sum of Quantity	30K	30,348	Yes
Sum of Sales	2.35M	2,348,361	Yes
Sum of Discount	885.55	885.55	Yes
Pie: Technology / Office Supplies / Furniture	886K / 824K / 639K	886,015 / 823,537 / 638,809	Yes
Pie share Technology	37.73%	37.73%	Yes
Line 2020: Technology / Office / Furniture	297K / 257K / 201K	296,989 / 257,218 / 200,823	Yes
Top product: Nokia Smart Phone, Full Size	31K	30,645	Yes
Country sales: France / Germany / UK	0.61M / 0.49M / 0.42M	0.61M / 0.49M / 0.42M	Yes

One card needs a note. The "Sum of Discount" card shows 885.55, but this is the sum of discount rates added across 8,045 lines (each between 0 and 0.8), so it has no business meaning. The average discount rate, 11.0%, is the useful figure. Section 9 suggests a replacement.

7. Data Analysis

7.1 Overall performance

Table 7: Headline measures

Measure	Value
Total sales	2,348,361
Total profit	283,202
Profit margin	12.1%
Units sold	30,348
Average discount rate	11.0%
Average order value	570
Average product-line value	292
Lines that lost money	1,734 (21.6%)
Total loss on those lines	-160,984

The business earns 12.1% on sales, but 21.6% of product lines lose money, and together those lines lose 160,984. That is more than half of the total profit of 283,202. Understanding where those losses come from is the key to improving profitability, and it is the focus of Sections 7.5, 7.6 and 7.9.

7.2 Trend over time

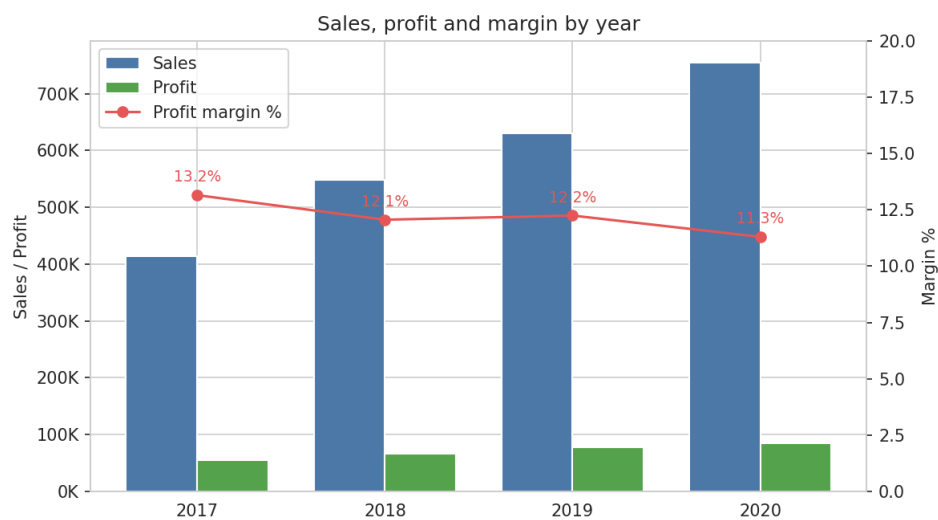


Figure 2: Sales, profit and profit margin by year

Table 8: Yearly performance

Year	Sales	Growth	Profit	Margin	Units
2017	414,493	-	54,514	13.2%	5,494
2018	548,698	32.4%	66,186	12.1%	7,086
2019	630,140	14.8%	77,172	12.2%	7,935
2020	755,030	19.8%	85,330	11.3%	9,833

Sales rose every year, from 414,493 in 2017 to 755,030 in 2020, an increase of 82% and a compound growth rate of 22.1% a year. Profit also rose, but only by 57%, so growth has not been fully converted into profit. The margin fell from 13.2% in 2017 to 11.3% in 2020. In other words, each additional unit of sales has been somewhat less profitable.

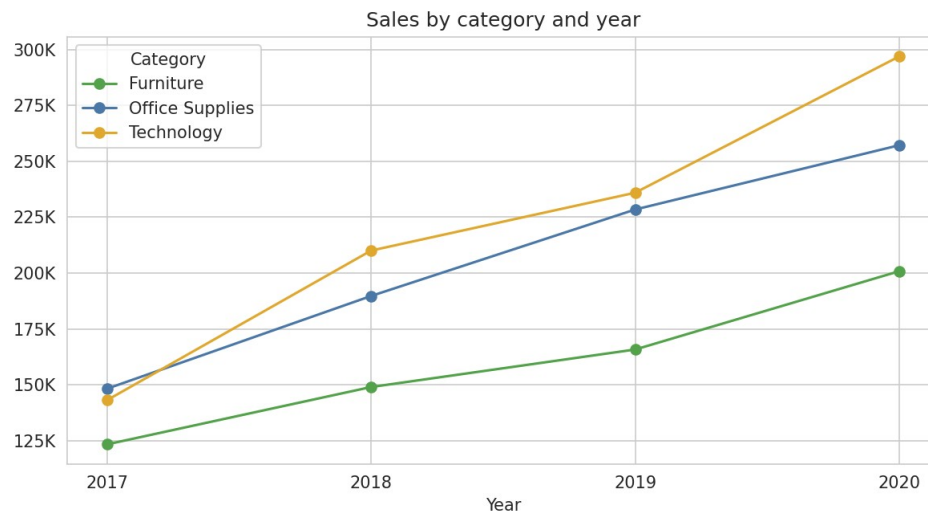


Figure 3: Sales by category and year (the Power BI line chart)

All three categories grew every year. Technology grew fastest (27.6% a year) and more than doubled, from 143,041 to 296,989. Office Supplies grew 20.2% a year and Furniture 17.7%. Technology overtook Office Supplies as the largest category in 2018 and has kept the lead since.

7.3 Seasonality

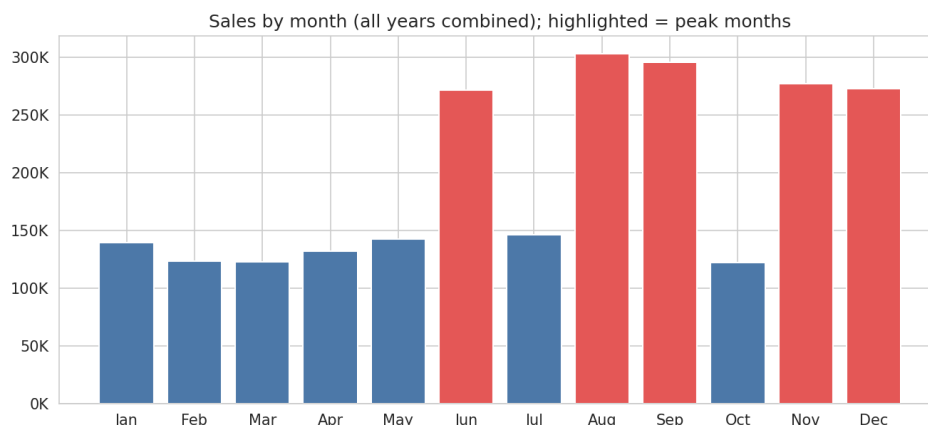


Figure 4: Sales by calendar month, all years combined

Sales are strongly seasonal. June, August, September, November and December each sell roughly twice as much as a typical month, and together they account for 60.5% of all sales. The other seven months are fairly flat at about 120K to 146K each. October is one of the weakest months despite sitting between two peaks. The file does not say why, so the cause (promotions, school and business calendars, or holidays) cannot be confirmed from this data. For planning, it means stock and staffing should be built up before the five peak months.

7.4 Category analysis



Figure 5: Share of sales, sales and profit, and margin by category

Table 9: Category performance

Category	Sales	Share	Profit	Margin	Avg discount
Furniture	638,809	27.2%	49,734	7.8%	15.5%
Office Supplies	823,537	35.1%	124,914	15.2%	9.5%
Technology	886,015	37.7%	108,554	12.3%	12.6%

Technology has the highest sales (886,015, 37.7%) but Office Supplies earns the most profit (124,914, 44.1% of the total) because its margin is the highest at 15.2% and its average discount the lowest at 9.5%. Furniture is the weakest category: 27.2% of sales but only 17.6% of profit, a margin of 7.8%, and it carries the highest average discount (15.5%). Office Supplies also dominates volume: 19,896 of 30,348 units sold.

7.5 Sub-category analysis

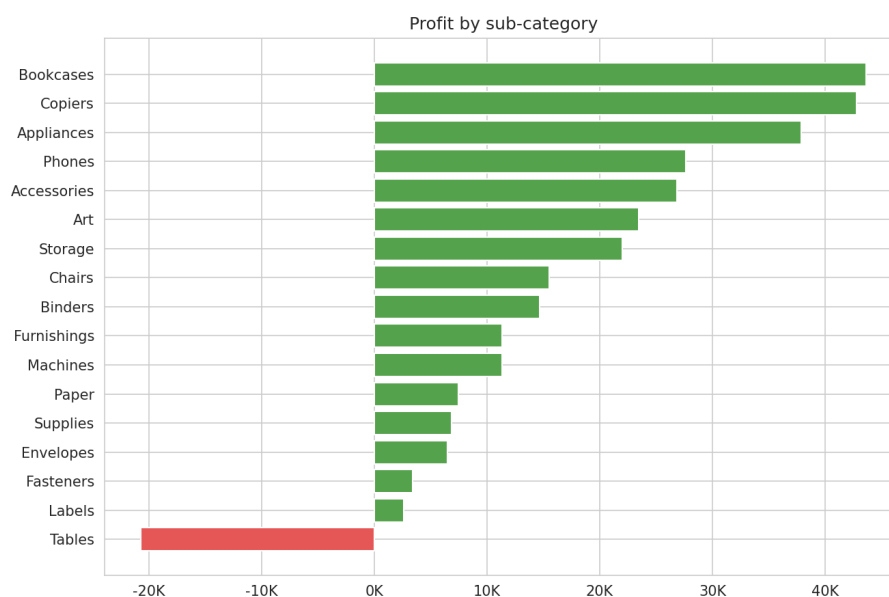


Figure 6: Profit by sub-category

Tables is the only loss-making sub-category, losing 20,731 on sales of 89,478. It has an average discount of 36.9%, and 83% of its lines carry a discount of 30% or more. Those lines lose 25,384, while the remaining Tables lines earn 4,653. Discounting, rather than the product itself, explains the loss.

Several big sub-categories earn thin margins: Machines 6.2% on 182,066 of sales, Storage 8.1%, Chairs 8.3% and Phones 9.8%. At the other end, small Office Supplies items (Labels, Fasteners, Envelopes, Paper) and Technology Accessories earn margins of about 20 to 21%, though on smaller sales. Bookcases (43,655) and Copiers (42,775) are the biggest profit contributors.

7.6 Country analysis

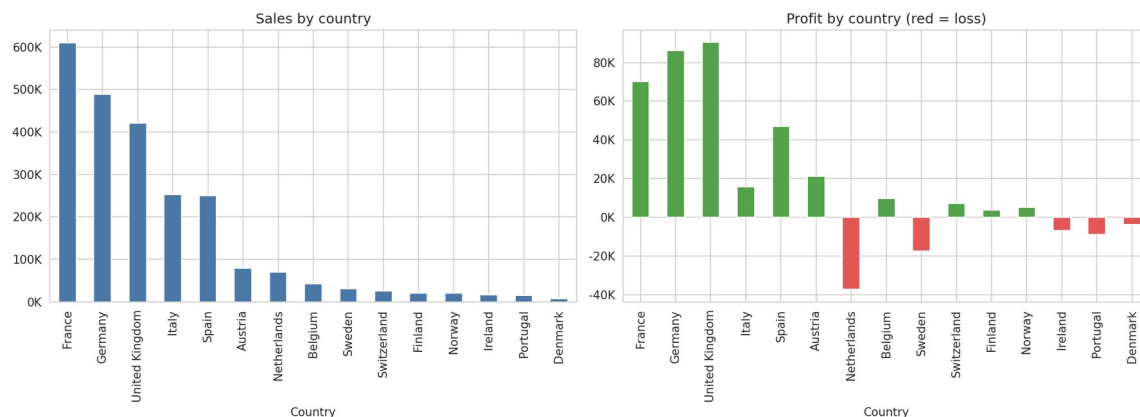


Figure 7: Sales and profit by country

Table 10: Country performance (sorted by sales)

Country	Sales	Profit	Margin	Avg discount
France	609,683	70,067	11.5%	7.7%
Germany	488,764	86,291	17.7%	5.5%
United Kingdom	420,414	90,370	21.5%	7.0%
Italy	252,742	15,802	6.3%	12.5%
Spain	249,318	47,039	18.9%	3.4%
Austria	79,382	21,332	26.9%	0.0%
Netherlands	70,313	-37,188	-52.9%	48.1%
Belgium	42,283	9,902	23.4%	0.0%
Sweden	30,490	-17,524	-57.5%	50.8%
Switzerland	24,874	7,234	29.1%	0.0%
Finland	20,702	3,908	18.9%	0.0%
Norway	20,529	5,167	25.2%	0.0%
Ireland	15,998	-6,886	-43.0%	50.3%
Portugal	15,106	-8,704	-57.6%	50.0%
Denmark	7,763	-3,608	-46.5%	50.8%

Sales are concentrated: France, Germany and the United Kingdom make up 64.7% of sales, and the top five countries (adding Italy and Spain) make up 86.1%. France is the largest market (609,683) but has a

modest margin of 11.5%. The United Kingdom (21.5%) and Germany (17.7%) are more profitable per unit of sales, and the UK is the largest source of profit (90,370). Italy is a concern: 252,742 of sales but only 6.3% margin.

5 countries lose money: Netherlands, Sweden, Ireland, Portugal and Denmark. Together they bring 139,670 of sales (5.9%) and a combined loss of 73,910. Their margins range from -58% to -43%. The dashboard hides most of this: its country chart is sorted by sales and has a scroll bar, so in the screenshot only the Netherlands is visible among the loss-making countries, while Sweden, Ireland, Portugal and Denmark sit off-screen at the far end.

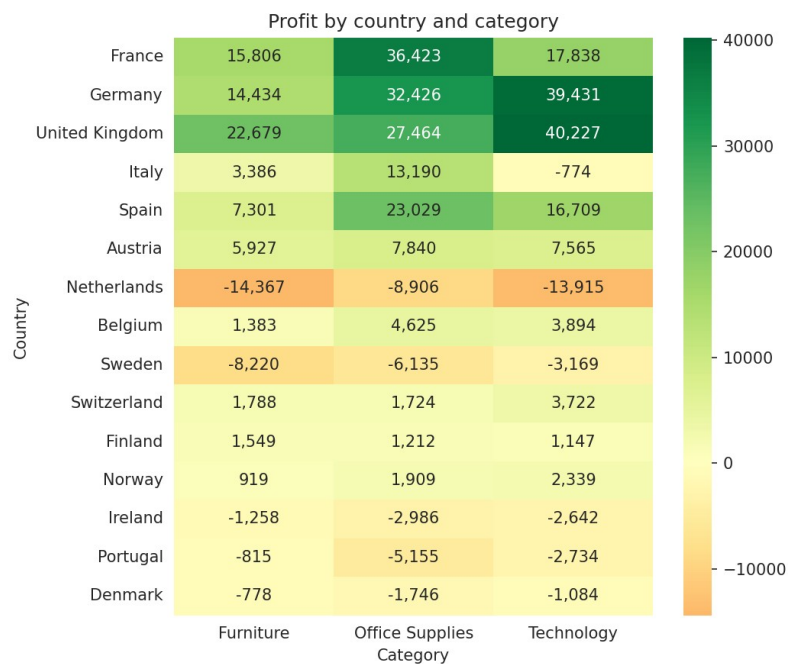


Figure 8: Profit by country and category

The heat map shows that the losses are not confined to one category. In the five loss-making countries, all three categories tend to lose money, which points to something that applies to the whole market, such as the pricing or discount policy, and not to a single product group. The next sections test this.

7.7 Region, customer segment and ship mode



Figure 9: Sales and profit by region, customer segment and ship mode

Table 11: Region, segment and ship mode

Group	Sales	Share of sales	Profit	Margin
Region: Central	1,315,299	56.0%	157,638	12.0%

Group	Sales	Share of sales	Profit	Margin
Region: North	515,896	22.0%	71,427	13.8%
Region: South	517,166	22.0%	54,137	10.5%
Segment: Consumer	1,236,245	52.6%	147,437	11.9%
Segment: Corporate	738,199	31.4%	92,649	12.6%
Segment: Home Office	373,917	15.9%	43,116	11.5%
Ship: Economy	1,412,511	60.1%	178,631	12.6%
Ship: Economy Plus	483,965	20.6%	54,336	11.2%
Ship: Immediate	131,459	5.6%	17,596	13.4%
Ship: Priority	320,426	13.6%	32,639	10.2%

The Central region produces 56.0% of sales. The North region has the best margin (13.8%) and the South the lowest (10.5%). By customer segment, Consumers buy 52.6% of sales, with Corporate customers second and slightly more profitable per unit of sales (12.6% against 11.9%). Economy shipping handles 60.1% of sales. Priority shipping has the lowest margin (10.2%). Differences between segments and ship modes are small compared with the differences by country and discount.

7.8 Products

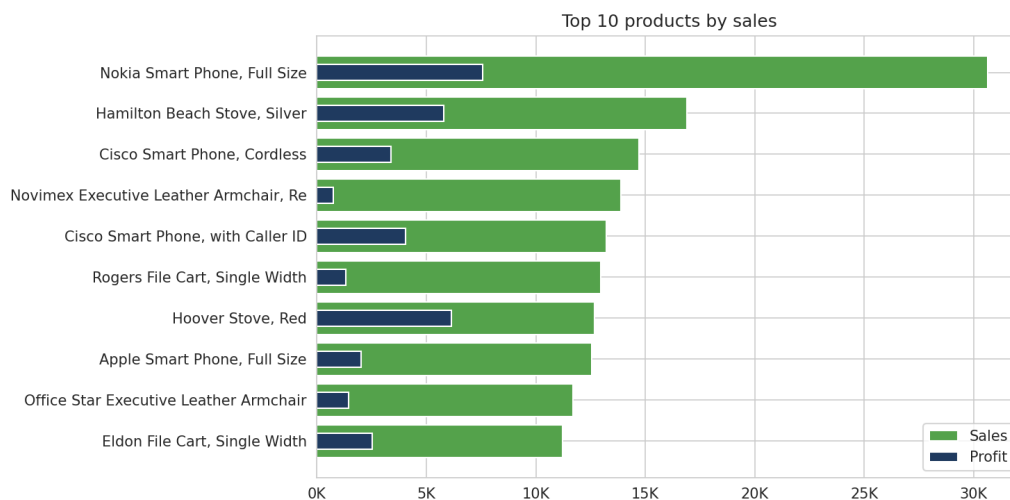


Figure 10: Top 10 products by sales (profit in the narrow bar)

The business sells 1,810 different products and the top ten make up only 6.4% of sales, so there is no heavy dependence on a few items. The best seller by far is the Nokia Smart Phone, Full Size, with 30,645 of sales and 7,583 of profit. The dashboard's "Top selling Product" chart shows the first five of these. Not all best sellers are equally profitable: the Novimex Executive Leather Armchair, Red sells 13,898 but earns only 736, and the Rogers File Cart sells 12,967 for a profit of 1,312.

The biggest loss-makers include Hoover Stove, White (-3,060), Office Star Executive Leather Armchair, Black (-2,323), Fellowes File Cart, Single Width (-2,267).

7.9 Discount analysis

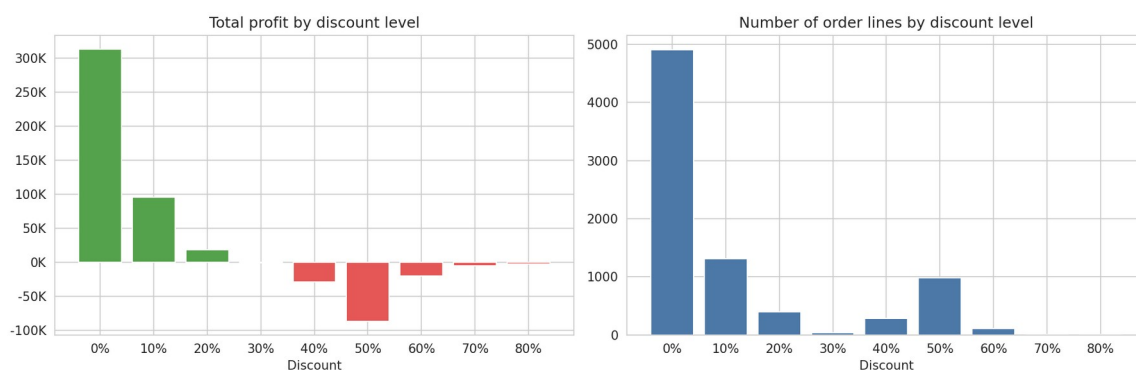


Figure 11: Total profit and number of lines by discount level

Table 12: Profit by discount band

Discount band	Lines	Sales	Profit	Margin
No discount	4,907	1,232,747	313,131	25.4%
10-20%	1,712	801,363	113,923	14.2%
30-40%	322	115,932	-28,233	-24.4%
50-80%	1,104	198,319	-115,619	-58.3%

Discount is the single strongest driver of profit in this data. Lines with no discount earn a margin of 25.4%, lines with 10 to 20% discount still earn 14.2%, but the margin turns negative at a discount of 30% and worsens sharply after that: -24.4% at 30 to 40% and -58.3% at 50 to 80%. The correlation between the discount rate and a line's profit margin is -0.82, a strong negative relationship.

Lines discounted by 30% or more are 1,426 of 8,045 (17.7%) and generate 314,251 of sales (13.4%), yet they lose 143,852. If these lines are set aside, the remaining 2,034,110 of sales earn 427,054 of profit, a margin of 21.0%, compared with 12.1% overall. The high-discount lines therefore wipe out about 34% of the profit earned everywhere else.

The geography of discounting explains the country results. The five loss-making countries all average a discount of about 48% to 51%, with 93% to 100% of their lines at 30% or more. In contrast, Austria, Belgium, Switzerland, Finland and Norway have no discounts at all and earn margins of 19% to 29%. In the Netherlands, 93% of lines are discounted by 30% or more; they lose 37,377, while the few remaining lines earn 189.

7.10 Returns

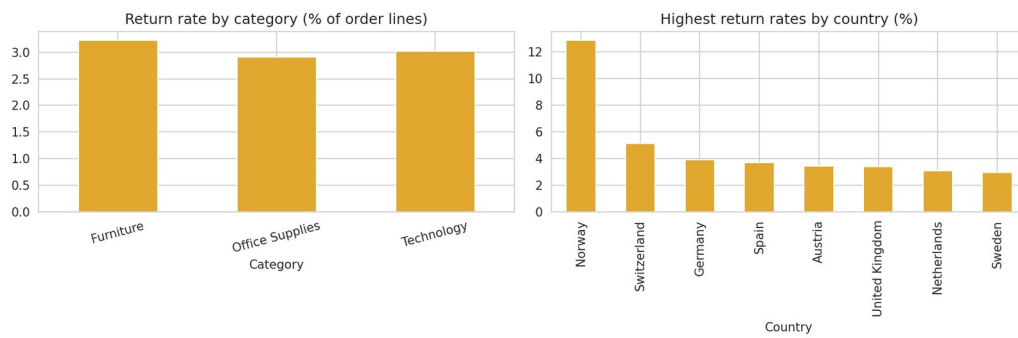


Figure 12: Return rate by category and by country

240 product lines (3.0%) were returned, worth 82,942 of sales (3.5%). Return rates are very similar across categories, from 2.9% to 3.2%, so returns are not a category problem. Norway shows the highest country rate (12.9%), but it rests on a very small number of lines (Norway has only 70 in total), so it should not be over-read.

A point of caution for the dashboard: returned lines still count in Sales (82,942) and still show a positive recorded profit (12,511). The dashboard totals therefore include sales that were later returned. Net sales excluding returns would be 2,265,419.

8. Summary of Key Insights

Table 13: Key insights

#	Insight	Evidence
1	Strong sales growth	Sales up 82% from 2017 to 2020 (22.1% a year)
2	Margin is shrinking	13.2% in 2017 to 11.3% in 2020
3	Technology leads sales; Office Supplies leads profit	Technology 37.7% of sales; Office Supplies 44.1% of profit
4	Furniture is the weakest category	Margin 7.8%; Tables lose 20,731
5	Sales are concentrated in a few markets	France, Germany, UK = 64.7% of sales
6	Five countries lose money	Netherlands, Sweden, Ireland, Portugal and Denmark: combined loss 73,910
7	Deep discounts cause the losses	Lines at 30% or more lose 143,852; other lines earn 21.0% margin
8	Strong seasonality	Five months bring 60.5% of sales
9	Returns are low and even	3.0% of lines in all categories
10	Dashboard figures are accurate	All values reconcile to the source data

9. Dashboard Evaluation and Improvements

9.1 Strengths

- Clear layout: a title, a block of four headline cards, a map and four chart panels read naturally from top to bottom.
- Consistent colour scheme and readable data labels.
- Linked slicers for Country, Sub-Category and Category make the page interactive.
- The line chart supports drill-down through the date hierarchy.
- All values are accurate (Section 6.4).

9.2 Issues and suggested improvements

Table 14: Dashboard review

Observation	Suggested change
The Discount card sums discount rates (885.55), which has no meaning.	Replace with Average Discount % (11.0%) or total discount amount.
No profit margin is shown, yet margin is the key health measure.	Add a Profit Margin % card (12.1%), and show margin by year and category.
The country chart is an area chart on a category axis and hides the last seven countries behind a scroll bar, including four of the five loss-making ones.	Use a sorted bar chart of profit by country, with negative values in red, so every country is visible.
The stacked column adds Profit on top of Sales, though Profit is part of Sales, so the stack total (sales + profit) is misleading.	Use a clustered column, or show Profit Margin % by category.
The pie chart and the column chart both show category sales.	Keep one and use the freed space for a sub-category profit chart (Figure 6).
The line chart shows sales only and there is no year filter.	Add a Year slicer and a profit or margin line.
Nothing on the page shows the discount problem found in Section 7.9.	Add a Profit by Discount Band chart and a Discount vs Margin visual.
Totals include returned items.	Add a Net Sales (excluding returns) measure.
The map sizes bubbles by sales only.	Colour bubbles by profit margin so loss-making countries stand out.

9.3 Suggested DAX measures

The following measures would support the changes above. They use the table and column names of the existing model.

```
Profit Margin % = DIVIDE(SUM('Order Breakdown'[Profit]), SUM('Order Breakdown'[Sales]))
Avg Discount % = AVERAGE('Order Breakdown'[Discount])
Net Sales = CALCULATE(SUM('Order Breakdown'[Sales]), 'Order Breakdown'[Return Status] = "Sold")
High Discount Profit = CALCULATE(SUM('Order Breakdown'[Profit]), 'Order Breakdown'[Discount] >= 0.3)
```

10. Business Recommendations

- **Control deep discounts.** Every discount level from 30% upward loses money in total. Set a maximum discount, or require approval above 30%, and track margin after discount.
- **Fix pricing in the five loss-making countries.** Netherlands, Sweden, Ireland, Portugal and Denmark sell at about 50% discount and lose 73,910. Moving them towards the discount policy of profitable markets would turn them profitable; if that is not possible, their continued presence needs review.
- **Reprice or reduce discounts on Tables.** The sub-category loses 20,731 only because of heavy discounting. Also review Machines, Storage and Chairs, whose margins are between 6% and 8%.
- **Look at Italy.** With 252,742 of sales and a 6.3% margin, its mix of discounts needs checking.
- **Grow the high-margin lines.** Office Supplies small items and Accessories earn about 20% margins. The no-discount markets (Austria, Belgium, Switzerland, Finland and Norway) earn 19% to 29% and are small today, so they are a growth opportunity.
- **Plan for seasonality.** Prepare stock and staff for June, August, September, November and December, which bring 60% of sales.
- **Watch margin, not only sales.** Margin fell from 13.2% to 11.3% while sales grew. Margin should be a headline measure on the dashboard.

11. Limitations

- **Discount and country are tied together.** Discounting appears to be applied market by market, so the effect of discount cannot be fully separated from the effect of country using this data alone.
- **No cost data.** Profit is taken as recorded, so the reasons for low margins (purchase cost, freight, promotions) cannot be examined.
- **Unknown currency.** The file does not state a currency, so amounts are shown without one.
- **Data-entry anomaly.** 24 orders have a ship date before the order date (Section 3.2).
- **Returns.** Return status is recorded but the date and reason of returns are not, and returned lines still carry profit.
- **Single dataset.** Findings describe this dataset and should be checked against more recent data before decisions are made.

12. Conclusion and Future Work

12.1 Conclusion

This project took a supermarket sales dataset through three stages: Excel pivot tables, a Power BI dashboard and a deeper Python analysis. The dashboard is accurate (every figure reconciles to the data) and gives a useful first view of sales by category, country and product. The deeper analysis showed that the business is growing quickly (22.1% a year) but its margin has fallen from 13.2% to 11.3%, and that the cause is mostly deep discounting: lines discounted by 30% or more lose 143,852, and the five markets that sell at about 50% discount all lose money.

The most valuable improvement is to bring this finding onto the dashboard, with a margin card, a profit-by-country chart that shows losses and a discount-versus-profit view, so that managers can see problems as soon as they appear.

12.2 Future work

- Add a Profit Margin card, Net Sales measure and Year slicer to the dashboard, and replace the area and stacked charts as described in Section 9.
- Add a second page for profit and discount analysis.
- Correct the ship-date anomalies and add delivery-time analysis by ship mode.
- Add cost data to separate product cost from discount effects.
- Build a simple forecast of monthly sales that uses the strong seasonal pattern.
- Publish the report to the Power BI service with scheduled data refresh.

Appendix A: Project Folder Structure

Sales_Data_Analysis_Project/	
data/European_supermarket_sales_data.xlsx	source data and Excel pivot tables
dashboard/bridge_group.pbix	Power BI dashboard file
dashboard/powerbi_dashboard_page1.png	dashboard screenshot
notebook/European_Sales_Analysis.ipynb	Python analysis with outputs
figures/	11 analysis charts
report/Sales_Data_Analysis_Report.docx	this report
report/results.json	numbers produced by the notebook

Appendix B: Key Python Code

```
orders = pd.read_excel(X, sheet_name='List of Orders')
lines = pd.read_excel(X, sheet_name='Order Breakdown')
df = lines.merge(orders, on='Order ID', how='left')
df['Year'] = df['Order Date'].dt.year

# headline KPIs (match the Power BI cards)
df.Sales.sum(), df.Profit.sum(), df.Quantity.sum(), df.Discount.sum()

# profit by discount level
df.groupby(df.Discount.round(1))[['Sales', 'Profit']].sum()
```